

Healthcare Appointment No-Show Prediction

Introduction

Healthcare appointment no-shows create significant challenges for hospitals and clinics by increasing operational costs, reducing efficiency, and affecting patient care. This project focuses on predicting whether patients are likely to miss their appointments using machine learning and data analytics techniques. The insights generated from the project can help healthcare organizations improve scheduling efficiency and patient engagement.

Abstract

The project analyzes appointment data to identify the major factors influencing patient no-shows. Data cleaning and preprocessing were performed using Python libraries such as Pandas and Scikit-learn. A Decision Tree classification model was developed to predict appointment attendance based on features like SMS reminders, patient age, appointment day, and medical conditions. Power BI dashboards were created to visualize trends and support decision-making. The project also provides optimization recommendations to reduce missed appointments and improve healthcare scheduling systems.

Tools Used

Category	Tools / Technologies
Programming Language	Python
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib
Machine Learning Model	Decision Tree Classifier
Visualization Tool	Power BI

Steps Involved in Building the Project

1. Data Collection: Imported healthcare appointment dataset for analysis.
2. Data Cleaning: Removed missing values, corrected data types, and handled duplicate records.
3. Feature Engineering: Created features such as weekday analysis, SMS reminder status, and patient age groups.
4. Exploratory Data Analysis (EDA): Analyzed appointment attendance trends across different categories.
5. Model Training: Built a Decision Tree classifier to predict patient no-shows.
6. Model Evaluation: Evaluated model performance using accuracy score, confusion matrix, precision, and recall.
7. Dashboard Development: Designed Power BI dashboards to visualize no-show patterns and appointment insights.
8. Optimization Recommendations: Suggested reminder systems, scheduling improvements, and patient engagement strategies to reduce missed appointments.

Project Deliverables

- Prediction model for appointment no-shows
- Power BI insight dashboard
- Optimization recommendations report
- Model performance metrics and confusion matrix

Conclusion

The Healthcare Appointment No-Show Prediction project successfully identified important factors affecting patient attendance and developed a predictive model for appointment management. The Power BI dashboards and machine learning insights can help healthcare providers improve operational efficiency, reduce revenue loss, and enhance patient care services. The recommendations generated from the project can support better scheduling strategies and improve overall healthcare management.